

JELIN RAPHAEL AKKARA

Summary: Pursuing PhD in Embodied AI. Experienced in computer vision, vision-language models, with research interests in failure recovery, memory, causal reasoning and representation learning.

Email: jelinraphael.akkara@phd.unipd.it

Portfolio: <https://jelinr.github.io/>

LinkedIn: [linkedin.com/in/jelin-raphael-akkara](https://www.linkedin.com/in/jelin-raphael-akkara)

GitHub: <https://github.com/JelinR>

SKILLS

Research Areas

Embodied AI, Computer Vision, Vision-Language Models, Multimodal Reasoning, Failure Recovery

Programming

Python, R, SQL

Machine Learning

PyTorch, TensorFlow, Keras, HuggingFace, Scikit-Learn, OpenCV, YOLO, Detectron2, Habitat AI, AI2THOR

Data & Scientific Computing

NumPy, Pandas, SciPy, Matplotlib, Seaborn, PySpark, Dask, Jupyter

Development

Git/GitHub, Linux, Bash, Docker, Hydra, SLURM

INITIATIVES

International Conference on Artificial Neural Networks ICANN 2026, Co-Organizer
14-17 September, 2026

Venice International Workshop, 'Physics of Data', Co-Organizer
23-24 May, 2024

CERTIFICATIONS

Language Proficiency in English (2022)
IELTS Band 8, CEFR Level C1

RESEARCH EXPERIENCE

GenAI Research Intern

ViWear

Enhanced Stable Diffusion-derived virtual try-on model by 7-12% (perceptual metrics), and lowered the inference time (1 fps to 3 fps). This was done using architectural improvements and fine-tuning.

Computer Vision Research Intern

Fondazione Bruno Kessler (FBK) / VIMP, University of Padova

Designed a modular framework for efficient Visual SLAM mapping (2D Semantic Map, VLFM). Developed a grid-based multi-modal embedding mapping, reducing memory usage by 99.75% relative to dense mappings.

EDUCATION

PhD in Brain, Mind and Computer Science (BMCS) 2025 - Present
University of Padua / Fondazione Bruno Kessler (FBK), Italy
Research Focus: Context-Aware, Adaptive Reasoning in Embodied AI

Masters in Physics of Data GPA : 110/110 2022 - 2025
University of Padua, Italy
Thesis: Zero-Shot Object Goal Navigation using Online Image Retrieval

B.Tech in Engineering Physics GPA : 8.43 / 10 2018 - 2022
National Institute of Technology, Calicut, India
First Class with Distinction

PUBLICATIONS

Akkara, J. R., Filippo Ziliotto, Serafini, L., Ballan, L., & Tommaso Campari. (2026). IMPRINT: Image-Conditioned Query Enrichment for Long-Tail Object Goal Navigation (2607.25106v1). arXiv. <https://arxiv.org/abs/2607.25106v1>

Ziliotto, F., **Akkara, J. R.**, Daniele, A., Ballan, L., Serafini, L., & Campari, T. (2026). Personal-3D: A Comprehensive Benchmark for Personalized Embodied AI Agents. International Journal of Computer Vision, 134(7). <https://doi.org/10.1007/s11263-026-02922-z>

Filippo Ziliotto, **Akkara, J. R.**, Daniele, A., Ballan, L., Serafini, L., & Tommaso Campari. (2025). PERSONAL: Towards a Comprehensive Benchmark for Personalized Embodied Agents (2509.19843v1). arXiv. <https://arxiv.org/abs/2509.19843v1>